

README and Guidance

Overview

This ReadMe file states how to replicate the data and results of the article *Fighting Crime in Lawless Areas: Evidence from Slums in Rio de Janeiro* (Bellégo C. and Drouard J.). The code in this replication package generates the data using Stata and Python, from five data sources, for the figures and tables in the article. The replicator should expect the code to run for about 5 hours.

Data Availability and Provenance Statements

Statement about Rights

We certify that the authors of the manuscript have legitimate access to and permission to use the data used in the manuscript.

We certify that the authors of the manuscript have documented permission to redistribute/publish the data contained within this replication package.

License for Data

The data and code are licensed under a Creative Commons/CC-BY-NC license.

Summary of Availability

All data are publicly available, however some of them are no longer available online. A copy of all the datasets can be found in the folder *data/source*.

Details on each Data Source

This section provides a detailed description of the original data. The original datasets come from five different sources.

1. Datasets from the Instituto de Segurança Pública (n.d.), the Institute that is in charge of recording crime data in the State of Rio de Janeiro.

- Crime data at district level can be downloaded from <http://www.ispdados.rj.gov.br/estatistica.html>, select "*Estatísticas de segurança: série histórica mensal por área de delegacia desde 01/2003*". A copy of the data, named *BaseDPEvolucaoMensalCisp.csv* (accessed March 2019), is provided as part of this archive. The data are in the public domain.
- Crime data at UPP level can be downloaded from <http://www.ispdados.rj.gov.br/UPP.html>, select "*Estatísticas de segurança: série histórica mensal por área de Unidade de Polícia Pacificadora (01/2007 a 06/2021)*". A copy of the data, named *UppEvolucaoMensalDeTitulos.csv* (accessed

March 2019), is provided as part of this archive. The data are in the public domain.

- Agregate time varying population in UPPs and in Rio de Janeiro can be downloaded from <http://www.ispdados.rj.gov.br/UPP.html>, select "*População: série anual projetada por UPP (2007 a 2021)*". A copy of the data, named `PopulacaoProjecaoUpp.xlsx` (accessed March 2021), is provided as part of this archive. The data are in the public domain.
- Calendar dates of bope intervention and of upp inauguration can be downloaded from <http://www.ispdados.rj.gov.br/UPP.html>, select "*UPP: datas de ocupação, instalação e extinção*". A copy of the data, named `UppDatasDeOcupacaoeInstalacao.xlsx` (accessed March 2021), is provided as part of this archive. The data are in the public domain.
- Layers of the districts of the State of Rio de Janeiro can be downloaded from <http://www.ispdados.rj.gov.br/Conteudo.html>, select "*Bases Cartográficas Digitais – Circunscrições Integradas de Segurança Pública (CISP)*". A copy of the data, named `Novo_Limite_CISP_WGS.shp` (accessed March 2019), is provided as part of this archive. The data are in the public domain.
- Layers of the UPPs of the State of Rio de Janeiro in folder can be downloaded from <http://www.ispdados.rj.gov.br/Conteudo.html>, select "*Bases Cartográficas Digitais - Unidades de Polícia Pacificadora (UPP)*". A copy of the data, `lm_upp_edit.shp` (accessed March 2019), is provided as part of this archive. The data are in the public domain.

2. Datasets from the Instituto Brasileiro de Geografia e Estatística (2010).

- Layers of the census tracts of the State of Rio de Janeiro can be downloaded from https://geoftp.ibge.gov.br/organizacao_do_territorio/malhas_territoriais/malhas_de_setores_censitarios__divisoes_intramunicipais/censo_2010/setores_censitarios_shp/rj/, select `rj_setores_censitarios.zip`. A copy of the data, named `33SEE250GC_SIR.shp` (accessed March 2020), is provided as part of this archive. The data are in the public domain.
- Socioeconomic characteristics of the census tracts of the State of Rio de Janeiro can be downloaded from https://ftp.ibge.gov.br/Censos/Censo_Demografico_2010/Sinopse/Agregados_por_Setores_Censitarios/, select `Base_informacoes_setores2010_sinopse_RJ.zip` or use directly the following link https://ftp.ibge.gov.br/Censos/Censo_Demografico_2010/Sinopse/Agregados_por_Setores_Censitarios/Base_informacoes_setores2010_sinopse_RJ.zip and the data will be downloaded automatically. A copy of the data, named `setores2010_sinopse_RJ.csv` (accessed March 2020), is provided as part of this archive. The data are in the public domain.

3. Datasets from the Prefeitura do Rio de Janeiro (n.d.). These datasets

have been collected from <https://www.data.rio/>, the open data portal of the municipality of Rio de Janeiro. This portal provides socioeconomic data from the 2010 Census, carried out by the Brazilian Institute of Geography and Statistics, and geospatial vector data at the census-tract level. This portal also provides geospatial vector data of favelas and UPPs of the city of Rio de Janeiro.

- Layers of the census tracts of the city of Rio de Janeiro with socioeconomic characteristics. The data were in the public domain but are no longer available online. A copy of the data, named `Razao_de_Sexo_HomensMulheres.geojson`, `Dom_Alugados.geojson`, `Inegabilidade.json`, `Dom_Coleta_de_Lixo.geojson`, `Densidade.json`, `Rendimen_nominal_mensal.json` (accessed June 2018) are provided as part of this archive.
- Layers of the UPPs of the city of Rio de Janeiro. The data were in the public domain but are no longer available online. A copy of the data, named `Limiteupp_arcgis.geojson` (accessed June 2018), is provided as part of this archive.
- Layers of the favelas of the city of Rio de Janeiro can be downloaded from https://www.data.rio/datasets/d8b4af0a02684c9a8408af2959fe73a0_4/about, click on *download* then select the preferred format. A copy of the data, named `Limite_Favelas_2010.geojson` (accessed June 2018), is provided as part of this archive. The data are in the public domain.
- Layers of the neighborhoods of the city of Rio de Janeiro can be downloaded from <https://www.data.rio/maps/limite-de-bairros>, click on *download* then select the preferred format. A copy of the data, named `Bairro.geojson` (accessed June 2018), is provided as part of this archive. The data are in the public domain.

4. Datasets from the Ministério da Saúde (n.d.).

- Data on the number of people who died in hospitals following attacks in the city of Rio de Janeiro, according to various characteristics, can be downloaded from http://tabnet.rio.rj.gov.br/cgi-bin/dh?sim/definicoes/sim_apos2005.def. From this page, choose the desired features (age, gender, year, type of aggression, etc.) and click on *"mostra"*. A html output page with a table containing the requested information is then generated and displayed. The data are in the public domain.

For our study, a multitude of queries were carried out by combining the possibilities for different categories on the race (category *Raça/Cor*), gender (category *Sexo*) and age (category *Faixa Etária*) of the victims and on the place (category *Local de Ocorrência*), the year (category *Ano do Óbito*) and the cause of death (category *Causa (CID10 3C)*). Copies of the html output pages (accessed July 2021) that were obtained are provided in a zip file as part of this archive. This zip file, named `SIM.zip`, contains 963 html

pages saved as text files, which correspond to the results of these various queries.

5. Dataset from the Shuttle Radar Topography Mission (Jarvis et al., 2008).

- Altitude data with high resolution can be downloaded from <https://srtm.csi.cgiar.org/srtmdata/>. Select the square on the map that corresponds to the area that contains Rio de Janeiro, choose "*Esri ASCII*" as format and " 5×5 " as tile size and then click on "*Search*". A new page is generated, with a description of the data. Click on "*Download SRTM*". A copy of the data, named `srtm_28_17.asc` (accessed July 2018), is provided as part of this archive. The data are in the public domain.

Table 1 summarizes the names of the source files, which were presented in more detail previously.

Table 1: Summary - data source

Information	Data file
Instituto de Seguranca Pública (n.d.)	
Crime data at district level	BaseDPEvolucaoMensalCisp.csv
Crime data at UPP level	UppEvolucaoMensalDeTitulos.csv
Agregate time varying population in UPPs	PopulacaoProjecaoUpp.xlsx
Dates of bope intervention and of upp inauguration	UppDadasDeOcupacaoInstalacao.xlsx
Layers of the districts of the State of RJ	Novo_Limite_CISP_WGS.shp
Layers of the UPPs of the State of RJ	lm_upp_edit.shp
Instituto Brasileiro de Geografia e Estatística (2010)	
<u>Census tracts of the State of RJ</u>	
Layers	33SEE250GC_SIR.shp
Socioeconomic characteristics	setores2010_sinopse_RJ.csv
Prefeitura do Rio de Janeiro (n.d.)	
Layers of the UPPs of the city of RJ	Limiteupp_arcgis.geojson
Layers of the favelas of the city of RJ	Limite_Favelas_2010.geojson
Layers of the neighborhoods of the city of RJ	Bairro.geojson
<u>Census tracts of the city of RJ</u>	
Layers and number of women/men by age category	Razao_de_Sexo_HomensMulheres.geojson
Layers and number owners/tenants	Dom_Alugados.geojson
Layers and inequality indicators	Inegabilidade.json
Layers and indicators on infrastructure/illiteracy/income	Dom_Coleta_de_Lixo.geojson
Layers and density/population	Densidade.json
Layers and household income	Rendimen_nominal_mensal.json
Ministério da Saúde (n.d.)	
Data on the number of people who died in hospitals following attacks in the city of RJ according to various characteristics	SIM.zip
Shuttle Radar Topography Mission (Jarvis et al., 2008)	
Altitude data with high resolution of RJ	srtm_28_17.asc

Dataset list

Preprocessed datasets A codebook named *documentation_of_datasets.xlsx* is provided in the replication package. This codebook provides all the information that is needed to understand the variables that can be found in the preprocessed datasets used to compute the figures and tables of the article.

Source datasets Descriptions or links pointing public codebooks that allow replicator to understand the variables that can be found in the source datasets (and that are used in our work) are provided below.

BaseDPEvolucaoMensalCisp.csv contains the monthly number of events per (crime) indicator registered in the districts of the state of Rio de Janeiro from 2007 to 2018. Codebook can be downloaded from <http://www.ispdados.rj.gov.br/Notas.html>, select "*Dicionário de variáveis: estatísticas de segurança por área de delegacia*".

UppEvolucaoMensalDeTitulos.csv contains the monthly number of events per (crime) indicator registered in UPPs of the state of Rio de Janeiro (and before the creation of the UPPs) from 2007 to 2018. Codebook can be downloaded from <http://www.ispdados.rj.gov.br/Notas.html>, select "*Dicionário de variáveis: estatísticas de segurança por UPP*".

PopulacaoProjecaoUpp.xlsx contains estimated population in all UPPs and in Rio de Janeiro by year (variables *38 UPP* and *Capital*).

Novo_Limite_CISP_WGS.shp provides the layers of the districts of the State of Rio de Janeiro. For each layer, the keys *geometry.coordinates* and *properties.DP* provide the coordinates of the spatial area and the id of the districts, respectively.

lm_upp_edit.shp provides the layers of the UPPs of the State of Rio de Janeiro. For each layer, the keys *geometry.coordinates* and *properties.nomeabrev* provide the coordinates of the spatial area and the name of the UPPs, respectively.

33SEE250GC_SIR.shp provides the layers of the census tracts of the State of Rio de Janeiro. For each layer, the keys *geometry.coordinates* and *properties.CD_GEOCOD* provide the coordinates of the spatial area and the id of the census tracts, respectively.

setores2010_sinopse_RJ.csv provides socioeconomic characteristics of the census tracts of the State of Rio de Janeiro. The variables *Cod_setor* and *V014* provide the id and the population in 2010 of the census tracts.

SIM.zip contains various html pages saved as text files with monthly number of people who died in hospitals following attacks according to various characteristics. The names of the different fields can be found on http://tabnet.rio.rj.gov.br/cgi-bin/dh?sim/definicoes/sim_apos2005.def, which is self-explanatory to understand the information returned by the different variables

srtm_28_17.asc contains altitude data with high resolution (spatial resolution approximately 250 meters) of a 5×5 degree zone which includes the city of Rio de Janeiro. The data contains the metadata to understand how to read it.

Limiteupp_arctgis.geojson provides layers of the UPPs of the city of Rio de Janeiro. For each layer, the keys *geometry.coordinates* and *properties.Nome* provide the coordinates of the spatial area and the name of the UPPs, respectively.

Limite_Favelas_2010.geojson provides layers of the favelas of the city of Rio de Janeiro. For each layer, the keys *geometry.coordinates* and *properties.cod_favela* provide the coordinates of the spatial area and the id of the favelas, respectively.

Bairro.geojson provides layers of the neighborhoods of the city of Rio de Janeiro. For each layer, the keys *geometry.coordinates* and *properties.Name* provide the coordinates of the spatial area and the id of the neighborhoods, respectively.

Datasets Razao_de_Sexo_HomensMulheres.geojson, Dom_Alugados.geojson, Inegabilidade.json, Densidade.json and Dom_Coleta_de_Lixo.geojson provide layers of the census tracts, with associated socio-economic characteristics, of the city of Rio de Janeiro. Tables 2-3 show the correspondence between the keys and their meaning.

Table 2: Correspondence between keys and meaning

key	meaning
Razao_de_Sexo_HomensMulheres.geojson	
<i>CDURPDBO_2</i>	id of the census tract
<i>CDURPDBO_9</i>	number of men
<i>CDURPDBO10</i>	number of men between 0-4 years
<i>CDURPDBO11</i>	number of men between 5-9 years
<i>CDURPDBO12</i>	number of men between 10-14 years
<i>CDURPDBO17</i>	number of men between 15-29 years
<i>CDURPDBO25</i>	number of men between 30-64 years
<i>CDURPDBO34</i>	number of men 65 years or more
<i>CDURPDBO35</i>	number of women
<i>CDURPDBO36</i>	number of women between 0-4 years
<i>CDURPDBO37</i>	number of women between 5-9 years
<i>CDURPDBO38</i>	number of women between 10-14 years
<i>CDURPDBO43</i>	number of women between 15-29 years
<i>CDURPDBO51</i>	number of women between 30-64 years
<i>CDURPDBO60</i>	number of women 65 years or more
Dom_Alugados.geojson	
<i>CDURPDBO_7</i>	id of the census tract
<i>CDURPDBO14</i>	number of owners
<i>CDURPDBO15</i>	number of tenants
<i>CDURPDBO16</i>	number of homes that are occupied
<i>CDURPDBO17</i>	number of homes that are neither owned/rented/occupied
Rendimento_nominal_mensal.json	
<i>CDURPDBO_2</i>	id of the census tract
<i>CDURPDBO_9</i>	average income per capita
<i>CDURPDBO10</i>	average income per home
Dom_Coleta_de_Lixo.geojson	
<i>BASEGEODBO</i>	id of the census tract
<i>BASEGEOD_1</i>	percentage of homes with drinking water
<i>BASEGEOD_2</i>	percentage of homes connected to sewer
<i>BASEGEOD_3</i>	percentage of homes connected to electricity
<i>BASEGEOD_4</i>	percentage of homes served by garbage collection
<i>BASEGEOD_5</i>	percentage of illiterate individuals between 10 and 14
<i>BASEGEOD_7</i>	percentage of illiterate individuals over 15
<i>BASEGEOD_8</i>	proportion of households with an income below the minimum wage
<i>BASEGEOD10</i>	proportion of households where the head of the household earns less than the minimum wage

Table 3: Correspondence between keys and meaning

key	meaning
Densidade.json	
<i>CDURP.DBO.Novos_Setores_Recortados_2010_1.ID_</i>	id of the census tract
<i>CDURP.DBO.Novos_Setores_Recortados_2010_1.Area</i>	surface area of the census tract
<i>CDURP.DBO.Dens_demo_.PopulacaoDPP</i>	population
<i>CDURP.DBO.Novos_Setores_Recortados_2010_1.Hab_ha</i>	density
<i>CDURP.DBO.Dens_demo_.DomParticularPerm</i>	number of households
<i>CDURP.DBO.Dens_demo_.MediaPessoasDPP</i>	average number of people living per home
Inegabilidade.json	
<i>CDURP.DBO.Novos_Setores_Recortados_2010_1.ID_</i>	id of the census tract
<i>CDURP.DBO.IDS_SETOR10.I_RENDARESP_P_MAI5DE10SM</i>	proportion of households where the head of the household earns more than 10 times the minimum wage

Computational requirements

Software Requirements

Python 3.9 and Stata 14 are used to build the datasets and compile the results.

Version of the packages.

- Python: *pandas 1.1.5*, *shapely 1.7.1*, *numpy 1.19.5*, *folium 0.13.0*, *csv 1.0*, *json 2.0.9*, *geopandas 0.11.1*, *geopy 2.2.0*.
- Stata: *estout* (as of 2017-05), *coefplot* (as of 2019-06), *did_multiplegt* (as of 2019-03), *boottest* (as of 2019-04), *asgen* (as of 2020-06)

Memory and Runtime Requirements

The code was run on a Intel(R) Core(TM) i7-6820HQ CPU @ 2.70GH with 16 GB of Random Access Memory and took approximately 5 hours to complete. The Python code took approximately 4 hours to complete using a Debian Operating System. The Stata code took about 1 hour to complete on Windows 10 operating system. Table 4 summarizes the estimated time to run each script.

Table 4: List of programs - Runtime

Script	Time to run
<u>Python</u>	
<i>post_aej_geodata1.py</i>	Less than 1 minute
<i>post_aej_geodata2.py</i>	Approximately 20 minutes
<i>post_aej_geodata3.py</i>	Less than 1 minute
<i>post_aej_socio_point.py</i>	Less than 1 minute
<i>post_aej_crimeDP.py</i>	Less than 1 minute
<i>post_aej_crimeUPP.py</i>	Less than 1 minute
<i>post_aej_dataUPP.py</i>	Approximately 12 minutes
<i>post_aej_geographical_analysis.py</i>	Approximately 3.5 hours
<i>post_aej_geographical_spillovers.py</i>	Approximately 5 minutes
<i>post_aej_datusus1.py</i>	Less than 1 minute
<i>post_aej_datusus2.py</i>	Less than 1 minute
<i>create_map.py</i>	Less than 1 minute
<u>Stata</u>	
<i>0_macro_libraries.do</i>	Less than 1 minute
<i>0_prog_eventStudy.do</i>	Less than 1 minute
<i>1_build_main_data.do</i>	Less than 1 minute
<i>2_descriptive_stat_and_main_analysis.do</i>	Approximately 1 hour
<i>3_analysis_event_study.do</i>	Approximately 2 minutes
<i>4_analysis_spillovers_outside_upps.do</i>	Less than 1 minute
<i>5_analysis_hospital.do</i>	Approximately 1 minute

Description of programs/code

The Python programs use raw datasets to create intermediary and preprocessed datasets, and also to produce the maps of the article. Then, the Stata programs use preprocessed datasets to create processed datasets and compute the figures and tables of the article.

Python programs

1. *create_data.py*

This program calls 11 modules that are in the subfolder *My_modules*.

a. *post_aej_geodata1.py*

This module creates *PrefeituraRio.json*, *UPPEstado_do_RioFinal.json* and *DPEstado_do_RioFinal.json* (which are saved in the folder *intermediary*).

- *PrefeituraRio.json* is data of the layers of census tracts of the city of Rio de Janeiro. For each census tract, we assign the values of different socioeconomic indicators. *Inegualdade.json*, *Razao_de_Sexo_HomensMulheres.geojson*, *Dom_Alugados.geojson*, *Densidade.json*, *Dom_Coleta_de_Lixo.geojson*, *Rendimen_nominal_mensal.json* are used to create *PrefeituraRio.json*.
- *UPPEstado_do_RioFinal.json* is roughly similar to *Limiteupp_arcgis.geojson*: one upp, which is outside the city of Rio de Janeiro, has been added.
- *DPEstado_do_RioFinal.json* is roughly similar to *Novo_Limite_CISP_WGS.shp*: some layers has been merged. This was done because police stations were opened during the period, and were thus combined with pre-existing police stations that were previously responsible for the jurisdiction of the new stations.

b. *post_aej_geodata2.py*

This module creates *PointCensusTract* (which is saved in the folder *intermediary*). *PointCensusTract* contains information about a grid of points of the city of Rio de Janeiro. Each point belongs to a census tract and has a 0.0005 degree space within each census tract. *PointCensusTract* is a list of three objects.

- (i) A Python dictionary that takes the id of census tracts as keys. Each census tract id returns a set of points. For each point we define to which district, which neighborhood, which favela (if it is the case), which upp (if it is the case) it belongs. In order to do so, we use the geospatial vector data *DPEstado_do_RioFinal.json*, *Bairro.geojson*, *Limite_Favelas_2010.geojson* and *UPPEstado_do_RioFinal.json*.

- (ii) A Python dictionary that takes the id of the points above mentioned. Each point id returns the localisation of the point (lon, lat).
- (iii) A Python dictionary that takes the id of the points above mentioned. Each point id returns the localisation of the point as `shapely.geometry.point.Point` type.

c. *post_aej_geodata3.py*

Using `PrefeituraRio.json` and `PointCensusTract`, this module creates `Valeur` (which is saved in the folder *intermediary*). This dataset, which is a Python dictionary, associates for each point of the grid the values of different socioeconomic indicators. For instance, if a census tract with P points has N inhabitants and $X\%$ of homeowner, we consider that each of these P points has a population of $\frac{N}{P}$ inhabitants and $X\%$ of homeowner.

d. *post_aej_socio_point.py*

This module creates `socio_favela_upp.csv` (which is saved in the folder *preprocessed*). This dataset associates for each point of the grid the values of different socioeconomic indicators, and indicates whether the point belongs to a favela and to a UPP. `PointCensusTract` and `Valeur` are used to build `socio_favela_upp.csv`.

e. *post_aej_crimeDP.py*

Using `UppEvolucaoMensalDeTitulos.csv`, this module creates `DP_Crime.csv` (which is saved in the folder *intermediary*). This dataset contains the monthly number of events per (crime) indicator registered in districts of the State of Rio de Janeiro from 2007 to June of 2016.

f. *post_aej_crimeUPP.py*

Using `BaseDPEvolucaoMensalCisp.csv`, this module creates `UPP_Crime.csv` (which is saved in the folder *preprocessed*). This dataset contains the monthly number of events per (crime) indicator registered in UPPs of the State of Rio de Janeiro from 2007 to June of 2016.

g. *post_aej_dataUPP.py*

This module creates `UPP_Data.csv` (which is saved in the folder *preprocessed*). This dataset contains all the main characteristics of the UPP of the city of Rio de Janeiro. To compute the socioeconomic characteristics at the UPP level, we use the grid of points of the city of Rio de Janeiro, with their characteristics, stored in `PointCensusTract` and `Valeur`. Dates of bope intervention and of upp inauguration come from `UppDatasDeOcupacaoInstalacao.xlsx`. Information about the criminal group that controlled each favela prior to the intervention

comes from several sources: Mapa do Ocupação Territorial Armada no Rio, favelascariocas.blogspot.com, InSight Crime, RioOnWatch, O Globo, Folha de S.Paulo. Information about UPP altitude characteristics is computed using Data srtm_28_17 and UPPEstado_do_RioFinal.json.

h. *post_aej_geographical_analysis.py*

Using `setores2010_sinopse_RJ.csv`, `33SEE250GC_SIR.shp`, `UPPEstado_do_RioFinal.json` and `PointCensusTract`, this module creates `CensusTractEstado_do_Rio.json`, `CensusTractMatrix` and `CensusTractPopPacified` (which are saved in the folder *intermediary*).

- `CensusTractEstado_do_Rio.json` contains the layers of census tracts of the State of Rio de Janeiro. For each census tract, we compute its centroid. Using their centroid, we define whether the census tracts lie in a UPP and to which district they belong.
- `CensusTractMatrix` is a list of four objects.
 - (i) A Python dictionary that takes the id of a census tract as key and returns the population of the census tract as value.
 - (ii) A Python dictionary that takes the id of a census tract as key and returns the coordinate of the centroid of the census tract as value.
 - (iii) A Python numpy object ($Number_census_tracts \times Number_census_tracts$) that returns the distances between all the census tracts of the State of rio de janeiro (distance=0 if between 0 and 100 meters; =1 if between 100 and 200 meters;...; =119 if between 11900 and 12000 meters; -1 if more than 12000 meters).
 - (iv) A Python numpy object ($Number_census_tracts \times 1$) that returns the date of pacification of the census tract when it lies inside a UPP (return 127 otherwise).
- `CensusTractPopPacified` is a list of two objects:
 - (i) A Python numpy object ($Number_census_tracts \times 120$) that returns, for each census tract, the size of the population living at the range distance of k meters.
 - (ii) A Python numpy object ($Number_census_tracts \times 120 \times 114$) that returns, for each census tract, the size of the population which is pacified at date t living at the range distance of k meters.

→ range distance k and date t are such that $k \in \{0 - 100, 100 - 200, ..., 11900 - 12000\}$ and $t \in \{1, ..., 114\}$ (which corresponds to the period January 2007 to June 2016).

i. *post_aej_spillovers.py*

Using `UPP_Crime.csv`, `DP_Crime.csv`, `CensusTractEstado_do_Rio.json`, `CensusTractPopPacified` and `CensusTractMatrix`, this module creates `CrimeDP_without_UPP.csv`, `pop_dp_no_upp.csv`, `Traitement_Denominateur_Spillovers.csv` and `Traitement_Numerateur_Spillovers.csv` (which are saved in the folder *preprocessed/spillovers*).

- (i) `CrimeDP_without_UPP.csv` contains the monthly number of events per (crime) indicator registered in districts (but not in UPPs) from 2007 to June of 2016. Events registered in UPPs have been removed from the events of this dataset.
 - (ii) `pop_dp_no_upp.csv` contains the number of individuals living in the district but not in the UPPs of the district.
 - (iii) `Traitement_Denominateur_Spillovers.csv` contains, for each district, the "average" number of individuals living within a range distance of k meters.
 - For a given district d , the "average" number of individuals living within a range distance k is defined as being $\sum_{c \in \Omega_d} \mu_{c,d} N_c^k$ with Ω_d being the set of census tracts that are inside district d but not in the area covered by the UPPs, $\mu_{c,d}$ the weight of census tract c in district d (*i.e.* the population of census tract c over the population of district d that lives outside the UPPs) and N_c^k the size of the population at the distance of k meters from census tract c .
 - (iv) `Traitement_Numerateur_Spillovers.csv` contains, for each district, the monthly "average" number of individuals living in a pacified area at date t within a range distance of k meters.
 - For a given district d , the "average" number of individuals living in a pacified area within a range distance k at date t is defined as being $\sum_{c \in \Omega_d} \mu_{c,d} n_{c,t}^k$ with Ω_d being the set of census tracts that are inside district d but not in the area covered by the UPPs, $\mu_{c,d}$ the weight of census tract c in district d (*i.e.* the population of census tract c over the population of district d that lives outside the UPPs) and $n_{c,t}^k$ the size of the population at the distance of k meters from census tract c that is pacified at date t .
- range distance k and date t are such that $k \in \{0 - 100, 100 - 200, \dots, 9000 - 9100\}$ and $t \in \{1, \dots, 114\}$ (which corresponds to the period January 2007 to June 2016).

j. `post_aej_datasus1.py`

Using `DATASUS/SIM` data, this module creates `agression.csv`, `agresion_arme_feu.csv` and `agresion_couteau.csv` (which are saved in the folder *preprocessed/hospital*).

- (i) `agression.csv` contains the recording of monthly deaths that occurred in an hospital of Rio de Janeiro following an attack (at the hospital level).

- (ii) `agression_arme_feu.csv` contains the recording of monthly deaths that occurred in an hospital of Rio de Janeiro following an attack with a firearm (at the hospital level).
- (iii) `agression_couteau.csv` contains the recording of monthly deaths that occurred in an hospital of Rio de Janeiro following an attack with a knife (at the hospital level).

k. *post_aej_datasus2.py*

Using `CensusTractEstado_do_Rio` and `CensusTractMatrix`, this module creates two datasets `Traitement_Denominateur_Hospital.csv` and `Traitement_Numerateur_Hospital.csv` (which are saved in the folder *preprocessed/hospital*).

- (i) `Traitement_Denominateur_Hospital.csv` contains, for each hospital, the number of individuals living within a range distance of k meters.
- (ii) `Traitement_Numerateur_Hospital.csv` contains, for each hospital, the monthly number of individuals living in a pacified area at date t within a range distance of k meters.

→ range distance k and date t are such that $k \in \{0 - 100, 100 - 200, \dots, 9000 - 9100\}$ and $t \in \{1, \dots, 114\}$ (which corresponds to the period January 2007 to June 2016). Only the 17 largest hospitals of the city of Rio de Janeiro are used to build these datasets.

2. *create_map.py*

Using `UPP_Data.csv`, `Limite_Favelas_2010.geojson`, `Limiteupp_arcgis.geojson` and `DPEstado_do_RioFinal.json`, this program creates the maps of the paper, that is: Figure 2, Figure B1 and Figure Q1. Note that this program creates the maps in an interactive html format. These files must then be manually adjusted to fit the paper version.

Stata programs

1. *0_macro_libraries.do*

This program sets up all the libraries to access the datasets used in the analysis. This program also install other Stata programs useful to produce some results.

Important: the user who want to replicate the results has to set his own root directory where the folder containing the code and data is located.

2. *0_prog_eventStudy.do*

This program contains the functions to estimate the event studies of the paper. It is called in the program *3_analysis_favelas_event_study.do*. This program does not have to be run.

3. *1_build_main_data.do*

This program generates processed data used to produce the main results of the paper.

4. *2_descriptive_stat_and_main_analysis.do*

This program produces from the datasets processed with the program *1_build_main_data.do* all the descriptive statistics of the paper and the main results of the paper, that is : Table 1, Table 2, Table 6 , Table 1A, Table A2, Figure 1, Figure A1, Figure B2, Figure B3, Figure P1, Table 3, Table 4, Table 5, Table 7, Figure 3, Table C1, Table E1, Table E2, Table F1, Table G1, Table I1, Table I2, Table J1, Table K1, Table K2, Table K3, Table L1, Table M1, Table M2, Table M3, Table M4, Table N1 and Table O1.

5. *3_analysis_event_study.do*

This program generates processed data and runs the regressions used to produce the event study results, that is : Figure 4 and Figure 5, Figure 6, Figure H1, Figure H2, Figure H3, Figure H4 and Figure P2.

The results of online appendix Figure H3 were produced using a version of the package *did_multipleGT* that is different from the last version of that package. The syntax used to propose the results does no longer work with the last version of the package. Therefore, we put the lines of code producing this Figure as a comment in the do file. An alternative code is also provided (still in comment), which produces results close to those presented in the paper but without generating event study figures.

6. *4_analysis_spillovers_outside_upps.do*

This program generates processed data and runs the regressions used to produce Table 8 (i.e., the results of the geographical analysis - spillovers outside favelas).

7.5 *analysis_hospital.do*

This program generates processed data and runs the regressions used to produce Table 9 (i.e., the results of the geographical analysis - victim profiles and killing methods).

Instructions to Replicators

To conduct replication, the following steps must be taken:

1. Folders *data*, *code* and *results* should be placed in the same folder
2. Folder *data* should contain a subfolder *source* with the raw datasets

3. Run *create_data.py*
4. Open the program *0_macro_libraries.do* and set the root directory or the root folder
5. Run the programs *0_macro_libraries.do*, *1_build_main_data.do*, *2_descriptive_stat_and_main_analysis.do*, *3_analysis_event_study.do*, *4_analysis_spillovers_outside_upps.do*, and *5_analysis_hospital.do*
6. Run the program *create_map.py*

Details

Folder *data* should contain a subfolder *source*. Subfolder *source* should contain 4 subfolders *ISP*, *IBGE*, *RioPrefeitura*, *STRM* and one file *SIM.zip*.

- Raw datasets *BaseDPEvolucaoMensalCisp.csv*, *UppEvolucaoMensalDeTitulos.csv*, *PopulacaoProjecaoUpp.xlsx*, *Novo_Limite_CISP_WGS.shp* and *lm_upp_edit.shp* must be stored in *ISP*
- Raw datasets *33SEE250GC_SIR.shp* and *2010_sinopse_RJ.csv* must be stored in *IBGE*
- Raw datasets *Limiteupp_arcgis.geojson*, *Limite_Favelas_2010.geojson*, *Bairro.geojson*, *Razao_de_Sexo_HomensMulheres.geojson*, *Inegabilidade.json*, *Dom_Alugados.geojson*, *Dom_Coleta_de_Lixo.geojson*, *Rendimen_nominal_mensal.json* and *Densidade.json* must be stored in *RioPrefeitura*
- Raw datasets *srtm_28_17.asc* must be stored in *STRM*

Folder *code* should contain a subfolder *My_modules*

- Programs *create_data.py*, *create_map.py*, *0_macro_libraries.do*, *0_prog_eventStudy.do*, *1_build_main_data.do*, *2_descriptive_stat_and_main_analysis.do*, *3_analysis_event_study.do*, *4_analysis_spillovers_outside_upps.do*, *5_analysis_hospital.do* must be stored in *code*
- Programs *post_aej_crimeDP.py*, *post_aej_crimeUPP.py*, *post_aej_datasus1.py*, *post_aej_datasus2.py*, *post_aej_dataUPP.py*, *post_aej_geodata1.py*, *post_aej_geodata2.py*, *post_aej_geodata3.py*, *post_aej_geographical_analysis.py*, *post_aej_geographical_spillovers.py*, *post_aej_socio_point.py* must be stored in *My_modules*

List of tables/figures and programs

Table 5-8 summarize the files produced, and by which program, to yield the figures and tables in the article.

Table 5: List of tables

Table	Program	No. Line	Output Files
Table 1	2_descriptive_stat_and_main_analysis.do	32	table1_col12.rtf
		34	table1_col34.rtf
Table 2	2_descriptive_stat_and_main_analysis.do	77	table2.xlsx
Table 3	2_descriptive_stat_and_main_analysis.do	217	table3_panelA.rtf
		224	table3_panelB.rtf
Table 4	2_descriptive_stat_and_main_analysis.do	297	table4.rtf
Table 5	2_descriptive_stat_and_main_analysis.do	312	table5.rtf
Table 6	2_descriptive_stat_and_main_analysis.do	86	table6.rtf
Table 7	2_descriptive_stat_and_main_analysis.do	326	table7.rtf
Table 8	4_analysis_spillovers_outside_upps.do	111	table8.rtf
Table 9	5_analysis_hospital.do	136	table9_panelA.rtf
		143	table9_panelB.rtf
		150	table9_panelC.rtf

Table 6: List of tables - online appendix

Table	Program	No. Line	Output Files
Table A1	2_descriptive_stat_and_main_analysis.do	18	tableA1.xls
Table A2	2_descriptive_stat_and_main_analysis.do	26	tableA2.rtf
Table C1	2_descriptive_stat_and_main_analysis.do	352	tableC1.rtf
Table E1	2_descriptive_stat_and_main_analysis.do	364	tableE1_panel1.rtf
		364	tableE1_panel2.rtf
		364	tableE1_panel3.rtf
Table E2	2_descriptive_stat_and_main_analysis.do	391	tableE2.rtf
Table F1	2_descriptive_stat_and_main_analysis.do	239	tableF1_panelA.rtf
		246	tableF1_panelB.rtf
Table G1	2_descriptive_stat_and_main_analysis.do	403	tableG1.rtf
Table I1	2_descriptive_stat_and_main_analysis.do	419	tableI1.rtf
Table I2	2_descriptive_stat_and_main_analysis.do	434	tableI2_panelA.rtf
		445	tableI2_panelB.rtf
Table J1	2_descriptive_stat_and_main_analysis.do	461	tableJ1.rtf
Table K1	2_descriptive_stat_and_main_analysis.do	488	tableK1.xlsx
Table K2	2_descriptive_stat_and_main_analysis.do	530	tableK2.rtf
Table K3	2_descriptive_stat_and_main_analysis.do	598	tableK3_final.xls
Table L1	2_descriptive_stat_and_main_analysis.do	633	tableL1.rtf
Table M1	2_descriptive_stat_and_main_analysis.do	652	tableM1_panelA.rtf
		659	tableM1_panelB.rtf
Table M2	2_descriptive_stat_and_main_analysis.do	673	tableM2_panelA.rtf
		680	tableM2_panelB.rtf
Table M3	2_descriptive_stat_and_main_analysis.do	691	tableM3_panelA.rtf
		698	tableM3_panelB.rtf
Table M4	2_descriptive_stat_and_main_analysis.do	709	tableM4.rtf
Table N1	2_descriptive_stat_and_main_analysis.do	721	tableN1.rtf
Table O1	2_descriptive_stat_and_main_analysis.do	736	tableO1_panelA.rtf
		743	tableO1_panelB.rtf

Table 7: List of figures

Figure	Program	No. Line	Output Files
Figure 1	2_descriptive_stat_and_main_analysis.do	137	figure1a_Murder_UPP_noUPP.pdf
		140	figure1b_Assault_UPP_noUPP.pdf
Figure 2	create_map.py	105	figure2.html
Figure 3	2_descriptive_stat_and_main_analysis.do	277	figure3.pdf
Figure 4	3_analysis_event_study.do	229	Figure4a_Murder.pdf
		250	Figure4b_Murder.pdf
		231	Figure4a_Assault.pdf
		252	Figure4b_Assault.pdf
		233	Figure4a_Robbery.pdf
		254	Figure4b_Robbery.pdf
		235	Figure4a_Theft.pdf
		256	Figure4b_Theft.pdf
		237	Figure4a_Extortion.pdf
		258	Figure4b_Extortion.pdf
		239	Figure4a_PoliceAction.pdf
		260	Figure4b_PoliceAction.pdf
		241	Figure4a_PoliceKill.pdf
		262	Figure4b_PoliceKill.pdf
Figure 5	3_analysis_event_study.do	243	Figure4a_Threat.pdf
		264	Figure4b_Threat.pdf
		245	Figure4a_Rape.pdf
		266	Figure4b_Rape.pdf
		247	Figure4a_TotalEvent.pdf
		268	Figure4b_TotalEvent.pdf
		274	Figure6a_Accident.pdf
		277	Figure6b_Accident.pdf

Table 8: List of figures - online appendix

Figure	Program	No. Line	Output Files
Figure A1	2_descriptive_stat_and_main_analysis.do	143	figureA1a_Policeaction_UPP_noUPP.pdf
		146	figureA1b_Policekill_UPP_noUPP.pdf
Figure B1	create_map.py	181	figureB1_august2010.html
		181	figureB1_december2010.html
		181	figureB1_july2012.html
		181	figureB1_june2008.html
		181	figureB1_june2011.html
		181	figureB1_june2014.html
		181	figureB1_march2009.html
		181	figureB1_september2013.html
Figure B2	2_descriptive_stat_and_main_analysis.do	157	figureB2a_population_timing.pdf
		160	figureB2b_density_pop_timing.pdf
		163	figureB2c_revenu_timing.pdf
		166	figureB2d_altitude_timing.pdf
Figure B3	2_descriptive_stat_and_main_analysis.do	169	figureB3a_elec_sewer_timing.pdf
		172	figureB3b_homeowner_young_literacy_timing.pdf
		175	figureB3c_infsmic_10xsmic_effpolice_timing.pdf
		178	figureB3d_murder_violence_timing.pdf
Figure H1	3_analysis_event_study.do	284	FigureH1a_Assault.pdf
		286	FigureH1a_Robbery.pdf
		288	FigureH1a_Theft.pdf
		290	FigureH1a_Threat.pdf
		293	FigureH1b_Assault.pdf
		295	FigureH1b_Robbery.pdf
		297	FigureH1b_Theft.pdf
		299	FigureH1b_Threat.pdf
Figure H2	3_analysis_event_study.do	306	FigureH2a_Extortion.pdf
		308	FigureH2a_Rape.pdf
		310	FigureH2a_TotalEvent.pdf
		313	FigureH2b_Extortion.pdf
		315	FigureH2b_Rape.pdf
		317	FigureH2b_TotalEvent.pdf
Figure H3	cannot be replicated		
Figure H4	3_analysis_event_study.do	375	FigureH4_NegBin_Murder.pdf
		377	FigureH4_NegBin_ViolentAssault.pdf
		379	FigureH4_NegBin_Robbery.pdf
		381	FigureH4_NegBin_Theft.pdf
		383	FigureH4_NegBin_PoliceAction.pdf
		385	FigureH4_NegBin_PoliceKill.pdf
		387	FigureH4_NegBin_Rape.pdf
		389	FigureH4_NegBin_Extortion.pdf
		391	FigureH4_NegBin_Threat.pdf
		393	FigureH4_NegBin_TotalEvent.pdf
Figure P1	2_descriptive_stat_and_main_analysis.do	197	figureP1a_police_actions.pdf
		199	figureP1b_firearm_confiscation.pdf
Figure P2	3_analysis_event_study.do	410	FigureP2_weapons.pdf
Figure Q1	create_map.py	222	figureQ1.html

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